

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

2026-2027- Odd Semester

III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Zeroth Review – 05.08.2026

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

FraudShield AI: Intelligent Digital Payment Risk Assessment System

Batch Members :

1.Barathraj M : 241901015

2.Deepan S A : 241901021

Supervisor:

Dr. S.Bhuvana

Assistant Professor[SG]

OUTLINE

- Abstract
- Objective
- Literature Survey
- Summary of Literature
- Proposed System & Novelty
- Block Diagram
- Hardware/Software requirements
- References

ABSTRACT

Context: With the rapid growth of online banking, UPI, mobile wallets, and card-based payments, accurate and real-time fraud prevention is crucial.

Core Technique: *FraudShield AI* is an AI-based decision support system utilizing **Knowledge Representation**,

Rule-Based Reasoning, and an **Inference Engine** running **Forward Chaining** over predefined IF-THEN rules.

Input Parameters: Evaluates payment amount, transaction time, device trust, geographic location, login behavior, transaction frequency, and account activity.

Risk Categorization: Classifies transactions into **Low, Medium, High, or Critical** risk levels.

Actionable Outputs: Provides real-time automated decisions—Approve, Request Additional Identity Verification, Temporarily Block, or Flag for Review.

Key Highlight: Incorporates Explainable AI (XAI) principles to provide transparent reasoning behind every risk decision, enhancing user trust and security scalability.

OBJECTIVE

- To develop an AI-based expert system for assessing risk in digital payment transactions.
- To represent financial payment knowledge using Predicate Logic and First Order Logic.
- To evaluate transaction patterns using rule matching algorithms.
- To filter low-risk transactions using parameter checks and behavioral baselines.
- To identify risk categories using Forward Chaining and explain them using rule-based reasoning traces.
- To generate a comprehensive transaction risk evaluation report with logical justification.

Literature Survey

TITLE	YEAR	AUTHOR	TECHNIQUE
Expert Systems: Principles and Programming	2005	Joseph C. Giarratano, Gary D. Riley .	Rule-Based Expert System, IF–THEN Rules
Artificial Intelligence: A Modern Approach	2021	Stuart Russell, Peter Norvig	Knowledge Representation, Inference Engine
Artificial Intelligence: A Guide to Intelligent Systems	2011	Micheal Negnevitsky	Knowledge-Based Systems, Rule Inference, Conflict Resolution, Explanation Facilities

EXISTING SYSTEM

- Payment risk assessments rely heavily on static monetary threshold limits.
- Risk attributes from different channels are analyzed separately, increasing assessment time.
- Rule updates depend largely on security teams' manual policy updates.
- Traditional systems primarily store payment records but do not provide intelligent reasoning.
- Handling complex multi-attribute payment fraud patterns is difficult.
- There is no automated multi-tier risk scoring or explainable decision support.
- Generating payment security audit logs requires significant manual effort.

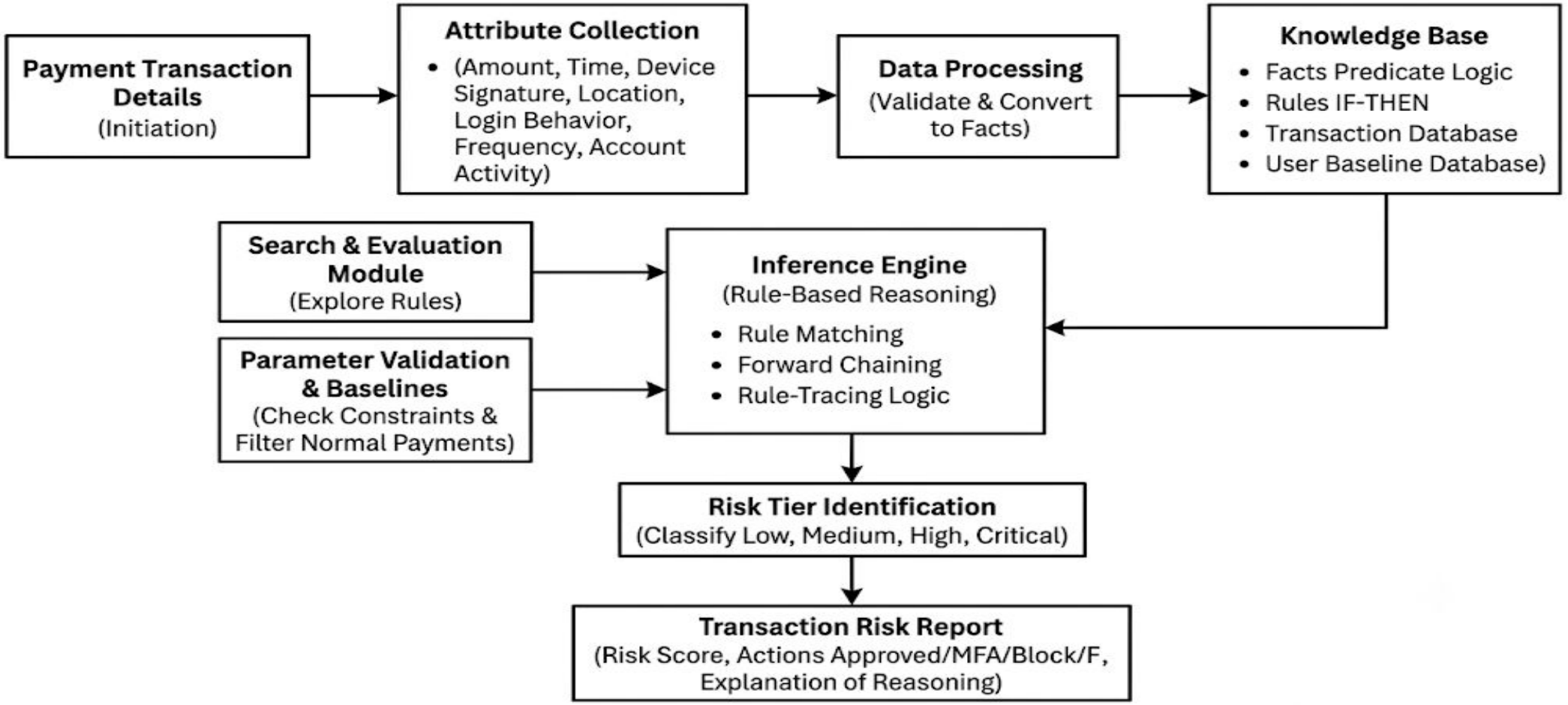
PROPOSED SYSTEM

- ➔ Utilizes structured payment transaction datasets from Kaggle along with user behavioral inputs.
- ➔ Stores transaction records, user baselines, device parameters, and login records in an SQLite database.
- ➔ Represents payment security knowledge using Predicate Logic and First-Order Logic.
- ➔ Uses rule evaluation routines to process transactional parameters.
- ➔ Applies parameter validation checks to filter standard payment activities.
- ➔ Applies Forward Chaining to assign Low, Medium, High, or Critical risk categories.
- ➔ Uses rule-tracing logic to explain the reasoning behind risk conclusions.
- ➔ Generates a Transaction Risk Evaluation Report.

Novelty in Proposed System

- Integrates multiple Classical AI techniques from the FOAI syllabus into a single application.
- Combines Rule Matchers, Parameter Validation, Production Rules, Forward Chaining, and Rule-Tracing Logic for intelligent risk assessment.
- Provides explainable decision support by tracing the logical reasoning behind every risk score.
- Represents payment facts and security risk conditions using Predicate Logic and First-Order Logic.
- Automatically filters normal transaction patterns using behavioral constraints before inference.
- Generates a detailed Transaction Risk Evaluation Report with the assigned risk level, triggered rules, reasoning, and recommendation.
- Designed as a modular rule-based expert system, allowing easy addition of new payment types, device metrics, and fraud detection rules without changing the core logic.

BLOCK DIAGRAM(PROPOSED SYSTEM)



Hardware/Software Requirements

Hardware Requirement:

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 500 MB free disk space
- Keyboard and Mouse
- Monitor (1366 × 768 resolution or higher)

Software Requirement

- Operating System: Windows 10/11
- Programming Language: Python 3.x
- IDE: Visual Studio Code / PyCharm / IDLE
- GUI Framework: Tkinter (if GUI is developed)
- Database: SQLite (optional, for storing transaction records)
- Web Browser: Google Chrome or Microsoft Edge (if web-based interface is used)

References

1. Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th Edition). Pearson.

1. Giarratano, J., & Riley, G. (2005). Expert Systems: Principles and Programming (4th Edition). Cengage Learning.

1. Negnevitsky, M. (2011). Artificial Intelligence: A Guide to Intelligent Systems (3rd Edition). Pearson Education.